

# Paper Directions 41 — my read (Sep 22, 2026)

Source: `transcripts/20260922_Paper_Directions_41_otter_ai_transcript.docx` (~8:29)  
Context: Charles walked Alex through a **cross-domain assortativity (H\_sort) screen** — Army, MBB, tenure, education, LoL, football (full roster + position groups).  
Prior: `PD30_notes.md` · Big Fish sandboxes: `../3-Master_Plan/re_entry/HEROs_and_PASSes/`

## Headline

**Priority flip (again): Assortativity is the blocker** — nail whether / how H\_sort matters before the story can move. **Education gets one more day** (college pond / prestige?), not a week. **Narrative restructure** (new PowerPoint aligned to story) by **end of week or weekend**.

- Alex (assortativity):** “Number one is still the biggest thing keeping us from moving forward... we got to nail that down.”
- Alex (education):** “I wouldn’t give up on the educational data... spend **a day**... that’s the biggest gain” among Big Fish if one can be massaged to work.
- Alex (academic decks):** “The problem is that **none of them work**” (HS pond → BA+ outcome).
- Alex (close):** By end of week: assortativity + education day **nailed** → then **story reshaped**.

Charles insight Alex accepted: Steep education HERO porches are **expected** — **congestion lives at college**, not high school; current Y is BA+ by follow-up, not selective advancement at the HS unit.  
Primary trio reaffirmed: **Army · MBB · Tenure** still valuable; Big Fish = probe / enrich, not replace.

## Transcript arc

| Block               | Time        | Content                                                                                                            |
|---------------------|-------------|--------------------------------------------------------------------------------------------------------------------|
| Army H_sort         | 00:00–00:43 | 0.255 → 0.279 after pool changes; Alex: H_sort delta “no difference”; hero porch <b>did</b> change materially      |
| MBB boundary        | 00:43–01:00 | H_sort ~0.064 — almost no assortativity; boundary-condition hypothesis                                             |
| Tenure · education  | 01:00–01:46 | Tenure ~0.121; HSB / NELs high H_sort; Charles: HS story ≠ congestion story                                        |
| LoL · football      | 01:46–02:57 | LoL ~0.445; FBS ~0.00 full roster; QB/WR/TE and DB groups “much better”                                            |
| Education fail mode | 02:57–03:48 | Alex: academic decks flat / wrong; missing <b>college attended</b> or <b>prestige</b> ?                            |
| Priority stack      | 03:50–07:38 | (1) assortativity, (2) education day, (3) narrative restructure; new datasets = ≤2 hr wrangle only                 |
| Luck / depth        | 06:15–06:55 | First three domains worked because of luck + deep domain knowledge (Army decades, tenure scrape)                   |
| Close / timeline    | 07:24–08:22 | End of week: assortativity + education; weekend: story PowerPoint; Charles showing decks = “exactly what I wanted” |

## H\_sort screen (as shown / discussed)

| Domain            | H_sort (PD41)      | Notes                                                            |
|-------------------|--------------------|------------------------------------------------------------------|
| Army              | ~0.255 → ~0.279    | Pool-grain / overlap rerun; porch shape changed more than H_sort |
| MBB               | ~0.064             | Very low — MBB as <b>low-sorting boundary</b>                    |
| Tenure            | ~0.121             | Respectable                                                      |
| HSB Soph          | high (Charles)     | Repo run Sep 22: <b>~0.284</b>                                   |
| NELS:88           | ~0.257 (Charles)   | Repo run Sep 22: <b>~0.251</b>                                   |
| LoL               | ~0.445             | Very high assortativity                                          |
| Football (FBS)    | ~0.00              | Full eligible cohort; position filters improve story             |
| Football QB/WR/TE | (not computed yet) | Old plot method — H_sort TBD                                     |
| Football DB       | (not computed yet) | ~15 per team — grouping by position                              |

Alex verdict — by domain (PD41)

| Domain                    | H_sort / porch                               | Alex line                                                         |
|---------------------------|----------------------------------------------|-------------------------------------------------------------------|
| Army                      | Strong; pool tweaks move porch               | “Very valuable” — scientific anchor                               |
| MBB                       | Low H_sort; familiar hero-ish curve          | Valuable as <b>no-assortativity</b> boundary                      |
| Tenure                    | Respectable H_sort                           | Valuable, “not as strong” as Army                                 |
| Education (HS pond → BA+) | High H_sort, steep HERO                      | “ <b>None of them work</b> ” for the story — wrong pond / wrong Y |
| LoL                       | Very high H_sort                             | Show metrics if appropriate; not lead domain                      |
| Football                  | ~0 full roster; usage/position groups better | Informational; position grouping matters                          |

Alex ask — priority stack (PD41 · by end of week)

| #  | Tier       | Task                          | Detail                                                                                         |
|----|------------|-------------------------------|------------------------------------------------------------------------------------------------|
| P0 | Primary    | Nail assortativity            | “Does assortativity matter?” — still main blocker to moving forward                            |
| P1 | Primary    | Education — one day           | Embed in data; check <b>college attended / prestige</b> (not just HS LOO → BA+); may fail — OK |
| P2 | Primary    | Narrative restructure         | New PowerPoint aligned to laid-out story — <b>end of week or weekend</b>                       |
| P3 | Optional   | New Big Fish datasets         | Only if $\leq$ ~2 hr download + GPT wrangle; Alex gave panels but no deep prep                 |
| P4 | Background | OpenAI / sports data partners | Charles exploring; low time investment                                                         |

Explicitly NOT this week: Rest of week on education; chasing Big Fish until one “wins”; deep new dataset builds without quick wrangle.

Education — open thread (Alex + Charles)

| Issue                       | Status                                                                             |
|-----------------------------|------------------------------------------------------------------------------------|
| Pond = high school          | Built — steep HERO expected (sorting + SES bundle)                                 |
| Y = BA+ at follow-up        | ~36% NELS / ~7% HSB Soph / ~17% HSB Senior — population rate, not top-K            |
| College attended / prestige | <b>Not dug in</b> — Alex thought this was planned; Charles to check CSV fields     |
| Congestion story            | Belongs at <b>college</b> pond, not HS — aligns with Charles + COMPASS read Sep 22 |

COMPASS read: PD41 **does not** reverse Sep 22 education sandbox work — it reframes **claim** (sorting contrast slide, not congestion replication). One-day Alex ask = **college-level** or prestige probe, not more HS ventile polish.

Delta vs PD30

| PD30                               | PD41                                                            |
|------------------------------------|-----------------------------------------------------------------|
| Primary = flipbook / story outline | Story still P0 but <b>blocked on assortativity</b>              |
| Big Fish = back burner             | <b>Education = one day</b> (biggest Big Fish upside if fixable) |
| Football usage = nugget            | Football + <b>position H_sort</b> = informational               |
| CAC / army access contingency      | Army pool grain + H_sort screen live in meeting                 |
| “What’s the story?”                | Same + <b>show me all domains on one assortativity axis</b>     |

Post-call repo work (same day — not in transcript)

Charles + COMPASS after the call (Sep 22):

- **NELS:88 + HSB Soph/Senior** 3x3 decks built via [scripts/big\\_fish\\_data\\_story.py](#)
- **Education primer** updated; synthesis table filled (H\_sort + steep HERO + “weak congestion”)
- **VECTOR manifest** updated for ChatGPT desktop repo onboarding (documentation only)

For next Alex touch: Cross-domain **H\_sort table** one-pager; education honest framing (HS sorting vs college congestion); Army porch after anchor-grain rerun.

Thread log

| Date       | Entry                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------------|
| 2026-09-22 | PD41 ingested; assortativity P0; education one-day + college prestige open; narrative restructure by EOW/weekend. |